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Question 1

Not answered

Marked out of 1

A case-control study is designed to explore the relationship between coeliac disease and schizophrenia. It is found that among 2000 cases who have schizophrenia, 3 have coeliac disease while among the 2000 controls 1 has coeliac disease. The odds ratio for exposure from this study is

Select one:

- ☐ 1
- ☐ 0
- ☐ 3
- ☐ 2
- ☐ -2

Your answer is incorrect.

The odds ratio is the cross-product ratio. The data must be redrawn as follows, before calculating the OR. $OR = (3 \times 1999) / (1 \times 1997) = 3$. Note that odds ratio cannot take a negative value. www.bmj.com/cgi/content/full/328/7437/438

exposure/outcome	schizophrenia	no schizophrenia	total
	cases	controls	
coeliac disease	3	1	4
no coeliac disease	1997	1999	3996
TOTAL	2000	2000	4000

The correct answer is: 3

Question 2

Not answered

Marked out of 1

A researcher is trying to construct a multiple linear regression analysis in a cohort study of risk factors for mortality in schizophrenia. He includes obesity, diabetes, suicide attempts and hyperlipidaemia in the regression analysis. He obtains unreliable results with respect to the significance of each of the above individual factors in predicting mortality. Which of the following is a correct explanation?

Select one:

- ☐ He has chosen risk factors that are unrelated to mortality in schizophrenia
- ☐ Regression analysis cannot be done using cohort studies
- ☐ Results are unreliable due to multiple testing
- ☐ The study was not blinded
- ☐ Misleading results are due to high correlation between some independent variables

Your answer is incorrect.

When two independent variables that are highly correlated are included in a regression equation, they will compete for statistical significance and may not appear as independent predictors of outcome. BMJ 2009; 338:b1038

The correct answer is: Misleading results are due to high correlation between some independent variables

Question 3

Not answered

Marked out of 1

In a small pilot study, 10 women with schizophrenia and 10 women with no psychosis were enquired regarding the habitual use of cannabis. The results are shown below (see image). What is the ratio of the odds that cannabis exposure is seen in schizophrenia patients to the odds that the exposure is seen in controls?

Matched subjects	Schizophrenia +	No psychosis
1	Cannabis user	Cannabis nonuser
2	Cannabis nonuser	Cannabis nonuser
3	Cannabis user	Cannabis user
4	Cannabis user	Cannabis user
5	Cannabis user	Cannabis nonuser
6	Cannabis nonuser	Cannabis nonuser
7	Cannabis user	Cannabis nonuser
8	Cannabis user	Cannabis nonuser
9	Cannabis nonuser	Cannabis user
10	Cannabis nonuser	Cannabis user

Select one:

- ☐ 1.25
- ☐ 6
- ☐ 3.33
- ☐ 2.25
- ☐ 3

Your answer is incorrect.

Odds ratio is the cross-product ratio. The data must be redrawn as follows, before calculating the OR. $OR = 36/16 = 2.25$

exposure/outcome ▼	schizophrenia ▼	no schizophrenia ▼	total ▼
	cases	controls	
cannabis use	6	4	10
no cannabis use	4	6	10
TOTAL	10	10	20

The correct answer is: 2.25



Question 4

Not answered

Marked out of 1

In a cohort study of PTSD, all cases that developed PTSD following exposure to a television programme were ascertained. If the relative risk for the association is equal to or less than 1.0 then

Select one:

- ☐ The study methodology has been flawed
- ☐ A valid comparison group cannot be obtained
- ☐ All subjects who were exposed to the TV programme will have PTSD
- ☐ There is no positive association between the television programme and PTSD
- ☐ All those who had PTSD in the sample were exposed to the TV programme

Your answer is incorrect.

If the relative risk is equal to 1, the numerator equals the denominator, and the risk in exposed persons equals the risk in nonexposed persons. Therefore, no support exists for an increased risk in exposed individuals or an association of the disease with the exposure in question.

The correct answer is: There is no positive association between the television programme and PTSD

Question 5

Not answered

Marked out of 1

The attributable risk of an exposure for a disease in a study is found to be 100%. This means

Select one:

- ☐ All those who are exposed will develop the disease
- ☐ All of the exposure in the population is attributable to the disease at 100%
- ☐ The disease will be completely eliminated from the population if the exposure is removed
- ☐ The chance of the disease occurring in the population is 100%
- ☐ All the incidence of the disease is the result of this single exposure

Your answer is incorrect.

If all the incidence of a disease were the result of a single factor, the attributable risk for that disease would be 100%. However, this is rarely if ever the case

The correct answer is: All the incidence of the disease is the result of this single exposure

Question 6

Not answered

Marked out of 1

An investigator examined cases of unplanned abortion in 3000 pregnant drug users. She classified mothers according to those who engaged with the deaddiction services and those who did not. She found that 11% of mothers who did not engage with the services had an abortion while only 3% of mothers who engaged with the local addictions services had an abortion. It was concluded that engagement with deaddiction during the months of pregnancy could substantially reduce fetal deaths due to abortion. This conclusion

Select one:

- ☐ May be incorrect because of low power of the study
- ☐ May be incorrect because mothers who engage with services during the month before childbirth may differ in other important characteristics from those who did not
- ☐ May be incorrect because prevalence rates are used where incidence rates are needed
- ☐ Is correct
- ☐ May be incorrect because there is no comparison group

Your answer is incorrect.

Prevalence rates are the only available data in case-control designs. This does not preclude making associations between variables. The more important problem here is the confounding due to other factors e.g. sound social support, better physical health, geographical residence, economic support, etc.

The correct answer is: May be incorrect because mothers who engage with services during the month before childbirth may differ in other important characteristics from those who did not

Question 7

Not answered

Marked out of 1

In a large case-control study of patients with anorexia nervosa, 19% of the patients were found to have a history of being abused as a child compared to 4% of a well-matched control group (matched by age, illness onset, ethnic group, and duration of illness). It was concluded that childhood abuse played a causal role in the anorexia nervosa. Which of the following is correct?

Select one:

- ☐ Lack of appropriate measurement for child abuse invalidates the results
- ☐ A clear dose response relationship has been demonstrated
- ☐ Lack of proper comparison group invalidates the result
- ☐ Effects of multiple mediators and confounders have not been adjusted
- ☐ The conclusion is correct

Your answer is incorrect.

Apart from lack of temporal correlation, one cannot exclude controlling for confounders in retrospective causal studies.

The correct answer is: Effects of multiple mediators and confounders have not been adjusted



Question 8

Not answered

Marked out of 1

In a study examining the relationship between domestic abuse and depression, women who do and do not suffer abuse are followed up over a five-year period. It is found that 70 of the 500 individuals who undergo domestic abuse develop depression; 150 of 3000 individuals who don't get abused develop depression. If the prevalence of domestic abuse in a particular council is 20%, the population attributable risk for depression is

Select one:

- ☐ 3.44
- ☐ 1.6
- ☐ 0.018
- ☐ 5.2
- ☐ 0.07

Your answer is incorrect.

Population attributable risk predicts the reduction in risk achievable if a risk factor is removed from a population. It is calculated by multiplying the attributable risk by the prevalence of exposure to the risk factor, and can be expressed by the formula $PAR = \text{Attributable risk} \times \text{Proportion of population exposed}$. Here, incidence in exposed is $70/500 = 14/100$. Incidence in nonexposed is $150/3000 = 5/100$. $AR = 9/100$. $PAR = 20\% \times 9/100 = 0.018$

The correct answer is: 0.018

Question 9

Not answered

Marked out of 1

Out of 100 consecutive depressed patients, 70 are smokers. Out of 100 consecutive non-depressed relatives accompanying a patient to the same clinic, 35 are smokers. How many times are the odds of being a smoker greater for a depressed patient than a non-depressed relative

Select one:

- ☐ 3.2 times
- ☐ 10 times
- ☐ 4.3 times
- ☐ 1.2 times
- ☐ 20 times

Your answer is incorrect.

Draw a table as below.

exposure/outcome	depression	no depression	total
	cases	controls	
smokers	70	35	105
non smokers	30	65	95
TOTAL	100	100	200

$$\text{Odds ratio} = (70 \times 65) / (30 \times 35) = 4.3$$

The correct answer is: 4.3 times

Question 10

Not answered

Marked out of 1

A case-control study compares low birth weight record of patients with schizophrenia and gender and race-matched controls. It concludes that the odds ratio of exposure is 5. The correct interpretation is

Select one:

- ☐ The birth weight of schizophrenia patients is 5 times lower than the birth weight off controls.
- ☐ Number of low birth weight subjects is 5 times more in schizophrenia group than control group
- ☐ Odds of having schizophrenia is 5 times greater for those who have low birth weight.
- ☐ Odds of having low birth weight is 5 times greater for those who have schizophrenia than those who do not.
- ☐ If low birth weight is completely eliminated, we will have a 5 fold reduction in incidence of schizophrenia

Your answer is incorrect.

The odds ratio, or the cross-products ratio, is the odds of an occurrence of interest in the study population, expressed as a proportion of the odds of the same occurrence in a control population. The correct interpretation in a case-control study must be that the odds of having low birth weight is five times higher in schizophrenia patients compared to controls.

The correct answer is: Odds of having low birth weight is 5 times greater for those who have schizophrenia than those who do not.

Question 11

Not answered

Marked out of 1

An environmental agency identifies 200 patients with life time history of panic attacks and 200 matched controls without the condition. Exposure to loud traffic noises was compared in two groups. 20 patients and ten controls had such an exposure. In the above study the case exposure rate is

Select one:

- ☐ (10/30)
- ☐ (10/200)
- ☐ (20/200)
- ☐ (20/30)
- ☐ (20/100)

Your answer is incorrect.

Case exposure rate is defined as $CER = (\text{no. of cases exposed to the exposure}) / (\text{total number of cases})$ $CER = 20 / 200$

The correct answer is: (20/200)



Question 12

Not answered

Marked out of 1

A group of 200 elderly depressed patients are followed up along with 200 age and gender-matched healthy controls for eight years to detect the development of dementia. Final data is shown in the below table. What is the attributable risk for depression?

	Dementia	No dementia	
Depression	25	175	200
No depression	10	190	200
	35	365	400

Select one:

- ☐ 1.25
- ☐ 7.5
- ☐ 2.5
- ☐ 0.125
- ☐ 0.075

Your answer is incorrect.

Attributable risk is the difference in risk between the exposed and the non-exposed. The risk is same as incidence rate. $AR = 25/200 - 10/200 = 15/200 = 0.075$.

The correct answer is: 0.075

Question 13

Not answered

Marked out of 1

An environmental agency identifies 200 patients with life time history of panic attacks and 200 matched controls without the condition. Exposure to loud traffic noises was compared in two groups. 20 patients and ten controls had such an exposure. In the above study the odds ratio of being exposed to loud noise in those who have panic attacks compared to those who do not is

Select one:

- ☐ $(20 \times 180) / (10 \times 190)$
- ☐ $(10 \times 190) / (20 \times 180)$
- ☐ $(20 \times 190) / (10 \times 180)$
- ☐ $(20 \times 10) / (400)$
- ☐ $(20 \times 200) / (10 \times 200)$

Your answer is incorrect.

To calculate odds ratio, one must construct a 2X2 table. See the attached image.

	Panic Disorder	No panic disorder	total
Loud noise exposed	20	10	30
Not exposed	180	190	370
total	200	200	400

Case exposure rate : $20/200$

Control exposure rate : $10/200$

From this table, the cross product (odds ratio) will be $(20 \times 190) / (10 \times 180)$

The correct answer is: $(20 \times 190) / (10 \times 180)$

Question **14**

Not answered

Marked out of 1

A scandalous tabloid reported that Asperger's syndrome may be more common in infants born to mothers who have irregular periods in the year immediately preceding the birth. A medical charity funds a study to investigate this report. 100 mothers of children with Asperger's syndrome were interviewed with respect to their menstrual history. 100 mothers of similar age and gender distributed normal children were also interviewed. If this study is to be believed with reports of an odds ratio of 2.02 for Asperger's syndrome in mothers with irregular periods before conception (95% confidence limits 1.06 to 2.56) which of the following is true

Select one:

- ☐ If 90% confidence limits were reported, these might cross 1
- ☐ The odds of being born to a mother with irregular periods preceding conception is two times higher for children with Asperger's syndrome than controls
- ☐ If more mothers are interviewed the 90% confidence limits will become wider
- ☐ p value for this odds ratio is likely to be >0.05
- ☐ The risk of having a child with Asperger's is doubled if the mother has irregular periods in 12 months preceding conception

Your answer is incorrect.

Increasing sample size will narrow down the confidence intervals. Similarly, reducing the degree of confidence attached (confidence levels) will also narrow down the range. As 95% limits do not cross 1, it is likely that the p-value is <0.05 . As this is a case-control study, only odds ratio - not relative risk - could be produced.

The correct answer is: The odds of being born to a mother with irregular periods preceding conception is two times higher for children with Asperger's syndrome than controls

Question 15

Not answered

Marked out of 1

An environmental agency identifies 200 patients with lifetime history of panic attacks and 200 matched controls without the condition. Exposure to loud traffic noises was compared in two groups. 20 patients and ten controls had such an exposure. Which of the following cannot be calculated from the above information?

Select one:

- ☐ The odds of exposure in cases
- ☐ Relative risk
- ☐ Case exposure rate
- ☐ Control exposure rate
- ☐ Odds ratio

Your answer is incorrect.

From a case-control study, one cannot calculate relative risk. The risk is calculated from incidence rate (new events/cases occurring). As retrospective design does not allow identification of new cases vs. old cases, one must resort to odds ratio as an estimate of likelihood.

The correct answer is: Relative risk



Question 16

Not answered

Marked out of 1

If the relative risk of suicide in depression is 10 and the prevalence of depression is 5% of population at any given time, then the proportion of suicides attributed to depression is nearly

Select one:

- ☐ 0.31
- ☐ 0.5
- ☐ 0.03
- ☐ 0.1
- ☐ 0.08

Your answer is incorrect.

To calculate the population attributable fraction (PAF), which may be interpreted as the proportion of suicides attributable to depression, we can use an equation described by Levin (Levin ML. The occurrence of lung cancer in man. *Acta Unio Internationalis Contra Cancrum* 1953;9:531-541). The equation is $PAF = \frac{Pe(RR - 1)}{1 + (Pe(RR - 1))}$, where Pe is the proportion of the population exposed to the factor (in this case, depression), and RR is the unadjusted relative risk of suicide associated with depression. This will yield $PAF = \frac{[(0.05)(9)]}{[1 + ((0.05)(9))]} = 0.45 / 1.45 = 31\%$

The correct answer is: 0.31

Question 17

Not answered

Marked out of 1

Among 100 smokers followed up until death 35 develop cognitive impairment. Among 100 non-smokers followed up until death, only 15 develop cognitive impairment. Risk value attributable to smoking is

Select one:

- ☐ 15/100
- ☐ 35/15
- ☐ $35 \times 15/100$
- ☐ 20/100
- ☐ 35/100

Your answer is incorrect.

Risk value attributable to exposure is the attributable risk. It is derived from obtaining the difference between control and experimental event rate. Here control event rate (no. of cognitive impairments among non-smokers) is 15/100. Experimental event rate (no. of cognitive impairments among smokers) is 35/100. $AR = EER - CER$ $AR = 35/100 - 15/100$ $AR = 20/100$

The correct answer is: 20/100

Question 18

Not answered

Marked out of 1

National Comorbidity Survey examined the possible association between traits of neuroticism and lifetime prevalence of various psychiatric disorders. Trained non-clinical staff screened randomly chosen individuals using well-validated questionnaires to establish lifetime diagnoses in addition to collecting data on drug use, alcohol intake and various personality traits. A sample of 200 subjects from the below survey show the following results : From the above study, the odds of being neurotic to being non-neurotic, when depressed, is

	Lifetime diagnosis of depression	No depression	
Neuroticism +	80	25	105
Neuroticism -	10	85	95
	90	110	200

Select one:

- ☐ 8
☐ 0.4
☐ 9.72
☐ 2.5
☐ 27.2

Your answer is incorrect.

The odds of having neuroticism when depressed is 80:10 = 8. Note that the question is not about Odds Ratio.

The correct answer is: 8

Question **19**

Not answered

Marked out of 1

Among 100 smokers followed up until death 35 develop cognitive impairment. Among 100 non-smokers followed up till death, only 15 develop cognitive impairment. If 25% of UK population are smokers, then the proportion of those with cognitive impairment attributable to smoking using the above study data is

Select one:

- ☐ 20/100
- ☐ 25/100
- ☐ 4/100
- ☐ 50/100
- ☐ 1/20

Your answer is incorrect.

Population attributable risk is defined as the proportion of cases that would not have occurred if the exposure is completely eliminated from the population. It is derived from multiplying ARR from a study data with the proportion of a population that is exposed. $PAR = ARR \times \text{proportion of population exposed}$. $PAR = 20/100 \times 25/100 = 1/20$

The correct answer is: 1/20

Question 20

Not answered

Marked out of 1

Which of the following cannot be established from the above study

Select one:

- ☐ Confidence intervals of odds ratio
- ☐ Dose-response relationship between neuroticism and depression
- ☐ Prevalence of lifetime depression
- ☐ Prevalence of lifetime neuroticism
- ☐ Temporal association between neuroticism and depression

Your answer is incorrect.

Case-control studies cannot establish a temporal association. But we can decipher dose response, prevalence rates and confidence intervals for the odds ratio.

The correct answer is: Temporal association between neuroticism and depression



Question **21**

Not answered

Marked out of 1

Please see the below table. What type of study is this could most possibly be?



Exposure	Cases	Controls	Univariate analysis	
	<i>n</i>	<i>n</i>	OR	(95% CI)
Schizophrenia diagnosis	100	110	1.73	(1.32-2.28)
Any <u>neuroleptic</u>	2246	1779	2.60	(2.43-2.78)
Any SSRI	955	892	2.01	(1.82-2.21)
Alcohol intake over recommended limit	243	288	1.60	(1.33-1.92)
Smoking status				
Current smoker	1304	1587	1.49	(1.37-1.62)
Ex-smoker	469	834	1.03	(0.91-1.16)

Select one:

- ☐ RCT
- ☐ Case Control
- ☐ Cohort study
- ☐ Cross sectional Survey

**Hormone replacement
therapy (women only)**

220

488

0.79

(0.66-0.94)

Your answer is incorrect.

Cases and controls are used; OR (not RR) is calculated - hence this must be a case control study.

The correct answer is: Case Control



Question **22**

Not answered

Marked out of 1

Please see the below table from a study investigating the risk factors of hip fracture using data from a primary care database. What was the most significant predictor of hip fractures from the given table?



Exposure	Cases	Controls	Univariate analysis	
	<i>n</i>	<i>n</i>	OR	(95% CI)
Schizophrenia diagnosis	100	110	1.73	(1.32-2.28)
Any <u>neuroleptic</u>	2246	1779	2.60	(2.43-2.78)
Any SSRI	955	892	2.01	(1.82-2.21)
Alcohol intake over recommended limit	243	288	1.60	(1.33-1.92)
Smoking status				
Current smoker	1304	1587	1.49	(1.37-1.62)
Ex-smoker	469	834	1.03	(0.91-1.16)

Select one:

- ☐ Neuroleptic use
- ☐ Alcohol use
- ☐ Smoking
- ☐ Schizophrenia
- ☐ Hormone replacement therapy

Hormone replacement therapy (women only)

220

488

0.79

(0.66-0.94)

Your answer is incorrect.

With high OR of 2.60, neuroleptic use seems to be the most significant predictor.

The correct answer is: Neuroleptic use



Question **23**

Not answered

Marked out of 1

Please see the below table. Which of the following predicts lesser hip fractures?



Exposure	Cases	Controls	Univariate analysis	
	<i>n</i>	<i>n</i>	OR	(95% CI)
Schizophrenia diagnosis	100	110	1.73	(1.32-2.28)
Any <u>neuroleptic</u>	2246	1779	2.60	(2.43-2.78)
Any SSRI	955	892	2.01	(1.82-2.21)
Alcohol intake over recommended limit	243	288	1.60	(1.33-1.92)
Smoking status				
Current smoker	1304	1587	1.49	(1.37-1.62)
Ex-smoker	469	834	1.03	(0.91-1.16)

Select one:

☐ Smoking☐ Schizophrenia☐ Hormone replacement therapy (women only)☐ Alcohol intake☐ Taking SSRIs

220	488	0.79	(0.66-0.94)
Hormone replacement therapy (women only)			

Your answer is incorrect.

With low OR of 0.79, Hormone replacement seems to predict lesser hip fractures (may be a positive / protective factor)

The correct answer is: Hormone replacement therapy



Question **24**

Not answered

Marked out of 1

Please see the below table. What can we NOT deduce from this study?



Exposure	Cases	Controls	Univariate analysis	
	<i>n</i>	<i>n</i>	OR	(95% CI)
Schizophrenia diagnosis	100	110	1.73	(1.32-2.28)
Any <u>neuroleptic</u>	2246	1779	2.60	(2.43-2.78)
Any SSRI	955	892	2.01	(1.82-2.21)
Alcohol intake over recommended limit	243	288	1.60	(1.33-1.92)
Smoking status				
Current smoker	1304	1587	1.49	(1.37-1.62)
Ex-smoker	469	834	1.03	(0.91-1.16)

Select one:

- ☐ Being treated with neuroleptics is significantly associated with hip fractures.
- ☐ Having smoked in the past is significantly associated with risk of hip fractures.
- ☐ Antipsychotics cause hip fractures.
- ☐ A diagnosis of schizophrenia is significantly associated with hip fractures.
- ☐ Hormone replacement is associated with lesser risk of hip fractures

Your answer is incorrect.

We cannot make causal inferences such as statement A in a case-control study. We can infer the strength of association between two variables (as in statements B, C, D and E). Using confidence intervals provided for OR, one can infer that having been a smoker in the past is not significantly associated with hip fractures - as the limit crosses 1.

The correct answer is: Antipsychotics cause hip fractures.

Question **25**

Not answered

Marked out of 1

Please see the below table. According to this study, the next best step for someone who has had a fall would be to:



Exposure	Cases	Controls	Univariate analysis	
	<i>n</i>	<i>n</i>	OR	(95% CI)
Schizophrenia diagnosis	100	110	1.73	(1.32-2.28)
Any <u>neuroleptic</u>	2246	1779	2.60	(2.43-2.78)
Any SSRI	955	892	2.01	(1.82-2.21)
Alcohol intake over recommended limit	243	288	1.60	(1.33-1.92)
Smoking status				
Current smoker	1304	1587	1.49	(1.37-1.62)
Ex-smoker	469	834	1.03	(0.91-1.16)

Select one:

- ☐ Do a thorough falls risk assessment
- ☐ Check weight
- ☐ None of the above
- ☐ Check for schizophrenia
- ☐ Check alcohol use

**Hormone replacement
therapy (women only)**

220

488

0.79

(0.66-0.94)

Your answer is incorrect.

No prescriptive clinical guidelines can be derived from this cross-sectional work. The results do not mean that every one with falls and hip fracture must be checked for missed schizophrenia or alcohol use!

The correct answer is: None of the above



Question **26**

Not answered

Marked out of 1

To determine the risk of 6 common cancers in patients with schizophrenia or bipolar disorder, data was obtained from 454 general practices contributing to the QRESEARCH general practice database. An open cohort of patients registered with these practices during the 10-year study period (January 1, 1995, to July 1, 2005) was included. Cases all involved patients aged between 25 and 100 years with a first-ever record of the index cancer during the study period. The sample consisted of 40 441 incident cases of 6 cancers (breast, colon, rectal, gastroesophageal, prostate, and respiratory). The authors found up to 5 controls per case matched by age, sex, general practice, and calendar time. Odds ratios were computed for cancer risk association with schizophrenia and bipolar disorder, after adjusting for smoking, body mass index, socioeconomic status, comorbidities, and prescribed medications, including antipsychotics. After adjustment, patients with schizophrenia had increased colon cancer risk (adjusted OR, 2.90; 95% confidence interval [CI], 1.85-4.57), a marginal increased breast cancer risk (adjusted OR, 1.52; 95% CI, 1.10-2.11), and a decreased respiratory cancer risk (adjusted OR, 0.53, 95% CI, 0.34-0.85). Patients with bipolar disorder had cancer risks similar to patients with neither condition after adjustment. The authors concluded that patients with schizophrenia had a significantly higher risk of colon cancer and a lower risk of respiratory cancer compared with control subjects. In contrast, the risks of cancer in patients with and without bipolar disorder are similar. What study design has been employed?

Table 1. Prevalence of Schizophrenia and Bipolar Disorder in Cases and Controls by Cancer Type

	Cancer					
	Breast	Colon	Rectal	Gastro-esophageal	Prostate	Respiratory
Cancer cases, No. (%)						
Schizophrenia only	47 (0.45)	31 (0.61)	8 (0.25)	12 (0.31)	12 (0.12)	20 (0.27)
Bipolar disorder only	42 (0.40)	12 (0.23)	6 (0.18)	7 (0.18)	15 (0.15)	32 (0.43)
Both conditions	2 (0.02)	1 (0.02)	0	2 (0.05)	1 (0.01)	3 (0.04)
Neither condition	10 444 (99.14)	5064 (99.14)	3234 (99.57)	3833 (99.46)	10 162 (99.73)	7451 (99.27)
Total	10 535	5108	3248	3854	10 190	7506
Cancer controls, No. (%)						
Schizophrenia only	143 (0.29)	48 (0.20)	48 (0.31)	48 (0.26)	108 (0.22)	139 (0.39)
Bipolar disorder only	154 (0.31)	55 (0.22)	30 (0.19)	36 (0.19)	85 (0.17)	94 (0.26)
Both conditions	10 (0.02)	3 (0.01)	2 (0.01)	5 (0.03)	8 (0.02)	9 (0.03)
Neither condition	49 767 (99.39)	24 352 (99.57)	15 472 (99.49)	18 388 (99.52)	48 547 (99.59)	35 739 (99.33)
Total	50 074	24 458	15 552	18 477	48 748	35 981

Select one:

- ☐ Simple case-control study
☐ Cross-sectional survey
☐ Prospective cohort study
☐ Retrospective cohort study
☐ Nested case-control study.

Your answer is incorrect.

This is a nested case-control study. Cases of cancer that occur in a defined cohort of GPRD-registered subjects are identified first. Then for each case, some matched non-cancer controls are selected from the same cohort.

The correct answer is: Nested case-control study.



Question **27**

Not answered

Marked out of 1

To determine the risk of 6 common cancers in patients with schizophrenia or bipolar disorder, data was obtained from 454 general practices contributing to the QRESEARCH general practice database. An open cohort of patients registered with these practices during the 10-year study period (January 1, 1995, to July 1, 2005) was included. Cases all involved patients aged between 25 and 100 years with a first-ever record of the index cancer during the study period. The sample consisted of 40 441 incident cases of 6 cancers (breast, colon, rectal, gastroesophageal, prostate, and respiratory). The authors found up to 5 controls per case matched by age, sex, general practice, and calendar time. Odds ratios were computed for cancer risk association with schizophrenia and bipolar disorder, after adjusting for smoking, body mass index, socioeconomic status, comorbidities, and prescribed medications, including antipsychotics. After adjustment, patients with schizophrenia had increased colon cancer risk (adjusted OR, 2.90; 95% confidence interval [CI], 1.85-4.57), a marginal increased breast cancer risk (adjusted OR, 1.52; 95% CI, 1.10-2.11), and a decreased respiratory cancer risk (adjusted OR, 0.53, 95% CI, 0.34-0.85). Patients with bipolar disorder had cancer risks similar to patients with neither condition after adjustment. The authors concluded that patients with schizophrenia had a significantly higher risk of colon cancer and a lower risk of respiratory cancer compared with control subjects. In contrast, the risks of cancer in patients with and without bipolar disorder are similar. Which of the following best describes the sampling procedure used for selecting controls?

Table 1. Prevalence of Schizophrenia and Bipolar Disorder in Cases and Controls by Cancer Type

	Cancer					
	Breast	Colon	Rectal	Gastro-esophageal	Prostate	Respiratory
Cancer cases, No. (%)						
Schizophrenia only	47 (0.45)	31 (0.61)	8 (0.25)	12 (0.31)	12 (0.12)	20 (0.27)
Bipolar disorder only	42 (0.40)	12 (0.23)	6 (0.18)	7 (0.18)	15 (0.15)	32 (0.43)
Both conditions	2 (0.02)	1 (0.02)	0	2 (0.05)	1 (0.01)	3 (0.04)
Neither condition	10 444 (99.14)	5064 (99.14)	3234 (99.57)	3833 (99.46)	10 162 (99.73)	7451 (99.27)
Total	10 535	5108	3248	3854	10 190	7506
Cancer controls, No. (%)						
Schizophrenia only	143 (0.29)	48 (0.20)	48 (0.31)	48 (0.26)	108 (0.22)	139 (0.39)
Bipolar disorder only	154 (0.31)	55 (0.22)	30 (0.19)	36 (0.19)	85 (0.17)	94 (0.26)
Both conditions	10 (0.02)	3 (0.01)	2 (0.01)	5 (0.03)	8 (0.02)	9 (0.03)
Neither condition	49 767 (99.39)	24 352 (99.57)	15 472 (99.49)	18 388 (99.52)	48 547 (99.59)	35 739 (99.33)
Total	50 074	24 458	15 552	18 477	48 748	35 981

Select one:

- ☐ Stratified sampling
- ☐ Incident density sampling
- ☐ Simple random sampling
- ☐ Systematic sampling
- ☐ Cluster based sampling

Your answer is incorrect.

Incidence density sampling is a technique used in nested case control studies to recruit controls within the cohort from where cases are derived. It involves matching each case to a set of individuals (usually >1) that are at risk of the exposure at the time of case occurrence. In other words, instead of selecting controls from any of the cancer-free individuals in the 10-year database, a systematic search (usually aided by a computer program) is undertaken to find cancer-free individuals matched with time at the time of case incidence. This time-matched sampling provides unbiased estimates of the risk of being exposed to the factor of interest (in this case – schizophrenia or bipolar disorder). <http://www.ncbi.nlm.nih.gov/pubmed/15550597>

The correct answer is: Incident density sampling



Question **28**

Not answered

Marked out of 1

To determine the risk of 6 common cancers in patients with schizophrenia or bipolar disorder, data was obtained from 454 general practices contributing to the QRESEARCH general practice database. An open cohort of patients registered with these practices during the 10-year study period (January 1, 1995, to July 1, 2005) was included. Cases all involved patients aged between 25 and 100 years with a first-ever record of the index cancer during the study period. The sample consisted of 40 441 incident cases of 6 cancers (breast, colon, rectal, gastroesophageal, prostate, and respiratory). The authors found up to 5 controls per case matched by age, sex, general practice, and calendar time. Odds ratios were computed for cancer risk association with schizophrenia and bipolar disorder, after adjusting for smoking, body mass index, socioeconomic status, comorbidities, and prescribed medications, including antipsychotics. After adjustment, patients with schizophrenia had increased colon cancer risk (adjusted OR, 2.90; 95% confidence interval [CI], 1.85-4.57), a marginal increased breast cancer risk (adjusted OR, 1.52; 95% CI, 1.10-2.11), and a decreased respiratory cancer risk (adjusted OR, 0.53, 95% CI, 0.34-0.85). Patients with bipolar disorder had cancer risks similar to patients with neither condition after adjustment. The authors concluded that patients with schizophrenia had a significantly higher risk of colon cancer and a lower risk of respiratory cancer compared with control subjects. In contrast, the risks of cancer in patients with and without bipolar disorder are similar. Which of the following biases is most likely to affect the validity of the results reported in this study?

Table 1. Prevalence of Schizophrenia and Bipolar Disorder in Cases and Controls by Cancer Type

	Cancer					
	Breast	Colon	Rectal	Gastro-esophageal	Prostate	Respiratory
Cancer cases, No. (%)						
Schizophrenia only	47 (0.45)	31 (0.61)	8 (0.25)	12 (0.31)	12 (0.12)	20 (0.27)
Bipolar disorder only	42 (0.40)	12 (0.23)	6 (0.18)	7 (0.18)	15 (0.15)	32 (0.43)
Both conditions	2 (0.02)	1 (0.02)	0	2 (0.05)	1 (0.01)	3 (0.04)
Neither condition	10 444 (99.14)	5064 (99.14)	3234 (99.57)	3833 (99.46)	10 162 (99.73)	7451 (99.27)
Total	10 535	5108	3248	3854	10 190	7506
Cancer controls, No. (%)						
Schizophrenia only	143 (0.29)	48 (0.20)	48 (0.31)	48 (0.26)	108 (0.22)	139 (0.39)
Bipolar disorder only	154 (0.31)	55 (0.22)	30 (0.19)	36 (0.19)	85 (0.17)	94 (0.26)
Both conditions	10 (0.02)	3 (0.01)	2 (0.01)	5 (0.03)	8 (0.02)	9 (0.03)
Neither condition	49 767 (99.39)	24 352 (99.57)	15 472 (99.49)	18 388 (99.52)	48 547 (99.59)	35 739 (99.33)
Total	50 074	24 458	15 552	18 477	48 748	35 981

Select one:

- ☐ Recall bias
- ☐ Attrition bias
- ☐ Information bias
- ☐ Selection bias
- ☐ Confounding bias

Your answer is incorrect.

In general, case-control studies use convenience sampling in which the controls are selected from advertisements or accessible volunteer pools. This can introduce selection bias as controls may be either 'super-controls' with no health problems or they may have a volunteer bias i.e. have some uncertain symptoms and approach a researcher to get easy access to lab tests/examination. These issues are obviated by using a nested design. This design also avoids recall bias as the exposure data is not obtained through recall. This design also reduces information bias as the data on risk factor exposure has been collected with investigator being blind to the case status at the time of recording the data. But given that nested designs are still observational designs, an association found between 2 variables could be explained by a third variable (confounder), which cannot be manipulated experimentally in this design.

<http://archpsyc.jamanetwork.com/article.aspx?articleid=482503>

The correct answer is: Confounding bias



Question **29**

Not answered

Marked out of 1

To determine the risk of 6 common cancers in patients with schizophrenia or bipolar disorder, data was obtained from 454 general practices contributing to the QRESEARCH general practice database. An open cohort of patients registered with these practices during the 10-year study period (January 1, 1995, to July 1, 2005) was included. Cases all involved patients aged between 25 and 100 years with a first-ever record of the index cancer during the study period. The sample consisted of 40 441 incident cases of 6 cancers (breast, colon, rectal, gastroesophageal, prostate, and respiratory). The authors found up to 5 controls per case matched by age, sex, general practice, and calendar time. Odds ratios were computed for cancer risk association with schizophrenia and bipolar disorder, after adjusting for smoking, body mass index, socioeconomic status, comorbidities, and prescribed medications, including antipsychotics. After adjustment, patients with schizophrenia had increased colon cancer risk (adjusted OR, 2.90; 95% confidence interval [CI], 1.85-4.57), a marginal increased breast cancer risk (adjusted OR, 1.52; 95% CI, 1.10-2.11), and a decreased respiratory cancer risk (adjusted OR, 0.53, 95% CI, 0.34-0.85). Patients with bipolar disorder had cancer risks similar to patients with neither condition after adjustment. The authors concluded that patients with schizophrenia have a significantly higher risk of colon cancer and a lower risk of respiratory cancer compared with control subjects. In contrast, the risks of cancer in patients with and without bipolar disorder are similar. Which of the following is true about this study design?

Table 1. Prevalence of Schizophrenia and Bipolar Disorder in Cases and Controls by Cancer Type

	Cancer					
	Breast	Colon	Rectal	Gastro-esophageal	Prostate	Respiratory
Cancer cases, No. (%)						
Schizophrenia only	47 (0.45)	31 (0.61)	8 (0.25)	12 (0.31)	12 (0.12)	20 (0.27)
Bipolar disorder only	42 (0.40)	12 (0.23)	6 (0.18)	7 (0.18)	15 (0.15)	32 (0.43)
Both conditions	2 (0.02)	1 (0.02)	0	2 (0.05)	1 (0.01)	3 (0.04)
Neither condition	10 444 (99.14)	5064 (99.14)	3234 (99.57)	3833 (99.46)	10 162 (99.73)	7451 (99.27)
Total	10 535	5108	3248	3854	10 190	7506
Cancer controls, No. (%)						
Schizophrenia only	143 (0.29)	48 (0.20)	48 (0.31)	48 (0.26)	108 (0.22)	139 (0.39)
Bipolar disorder only	154 (0.31)	55 (0.22)	30 (0.19)	36 (0.19)	85 (0.17)	94 (0.26)
Both conditions	10 (0.02)	3 (0.01)	2 (0.01)	5 (0.03)	8 (0.02)	9 (0.03)
Neither condition	49 767 (99.39)	24 352 (99.57)	15 472 (99.49)	18 388 (99.52)	48 547 (99.59)	35 739 (99.33)
Total	50 074	24 458	15 552	18 477	48 748	35 981

Select one:

- ☐ It is a retrospective design
- ☐ It is a secondary research design
- ☐ It is an experimental design
- ☐ It is a prospective design
- ☐ It is a qualitative design

Your answer is incorrect.

A nested case-control study is a retrospective observational study. It is a quantitative study but does not involve experimental manipulation. Temporal association between exposure and risk can be ascertained in nested case-control designs as the data though obtained retrospectively, has been recorded on the register (GPRD) prospectively.

The correct answer is: It is a retrospective design



Question 30

Not answered

Marked out of 1

To determine the risk of 6 common cancers in patients with schizophrenia or bipolar disorder, data was obtained from 454 general practices contributing to the QRESEARCH general practice database. An open cohort of patients registered with these practices during the 10-year study period (January 1, 1995, to July 1, 2005) was included. Cases all involved patients aged between 25 and 100 years with a first-ever record of the index cancer during the study period. The sample consisted of 40 441 incident cases of 6 cancers (breast, colon, rectal, gastroesophageal, prostate, and respiratory). The authors found up to 5 controls per case matched by age, sex, general practice, and calendar time. Odds ratios were computed for cancer risk association with schizophrenia and bipolar disorder, after adjusting for smoking, body mass index, socioeconomic status, comorbidities, and prescribed medications, including antipsychotics. After adjustment, patients with schizophrenia had increased colon cancer risk (adjusted OR, 2.90; 95% confidence interval [CI], 1.85-4.57), a marginal increased breast cancer risk (adjusted OR, 1.52; 95% CI, 1.10-2.11), and a decreased respiratory cancer risk (adjusted OR, 0.53, 95% CI, 0.34-0.85). Patients with bipolar disorder had cancer risks similar to patients with neither condition after adjustment. The authors concluded that patients with schizophrenia had a significantly higher risk of colon cancer and a lower risk of respiratory cancer compared with control subjects. In contrast, the risks of cancer in patients with and without bipolar disorder are similar. Which of the following is true with regard to the results reported in the table?

Table 1. Prevalence of Schizophrenia and Bipolar Disorder in Cases and Controls by Cancer Type

	Cancer					
	Breast	Colon	Rectal	Gastro-esophageal	Prostate	Respiratory
Cancer cases, No. (%)						
Schizophrenia only	47 (0.45)	31 (0.61)	8 (0.25)	12 (0.31)	12 (0.12)	20 (0.27)
Bipolar disorder only	42 (0.40)	12 (0.23)	6 (0.18)	7 (0.18)	15 (0.15)	32 (0.43)
Both conditions	2 (0.02)	1 (0.02)	0	2 (0.05)	1 (0.01)	3 (0.04)
Neither condition	10 444 (99.14)	5064 (99.14)	3234 (99.57)	3833 (99.46)	10 162 (99.73)	7451 (99.27)
Total	10 535	5108	3248	3854	10 190	7506
Cancer controls, No. (%)						
Schizophrenia only	143 (0.29)	48 (0.20)	48 (0.31)	48 (0.26)	108 (0.22)	139 (0.39)
Bipolar disorder only	154 (0.31)	55 (0.22)	30 (0.19)	36 (0.19)	85 (0.17)	94 (0.26)
Both conditions	10 (0.02)	3 (0.01)	2 (0.01)	5 (0.03)	8 (0.02)	9 (0.03)
Neither condition	49 767 (99.39)	24 352 (99.57)	15 472 (99.49)	18 388 (99.52)	48 547 (99.59)	35 739 (99.33)
Total	50 074	24 458	15 552	18 477	48 748	35 981

Select one:

- ☐ None of the rectal cancer cases had bipolar disorder
- ☐ The probability of schizophrenia is higher among prostate cancer cases than controls
- ☐ The probability of schizophrenia is higher among breast cancer cases than controls
- ☐ The probability of bipolar disorder is lower among respiratory cancer cases than controls
- ☐ 2% of patients with breast cancer had both schizophrenia and bipolar disorder

Your answer is incorrect.

It is important not to be carried away by the absolute numbers here, as there is an imbalance in sample size between the two groups. So to answer this MCQ, use the percentage values given in parentheses. From the given table, it is clear that the probability of schizophrenia is higher among breast cancer cases (0.45%) than controls (0.29%). The probability of schizophrenia is lower among prostate cancer cases (0.12%) than controls (0.22%). The probability of bipolar disorder is higher among respiratory cancer cases (0.43%) than controls (0.26%). Only 0.02% of patients with breast cancer had both schizophrenia and bipolar disorder. 6 of the rectal cancer cases had bipolar disorder. The correct answer is: The probability of schizophrenia is higher among breast cancer cases than controls



Question **31**

Not answered

Marked out of 1

To determine the risk of 6 common cancers in patients with schizophrenia or bipolar disorder, data was obtained from 454 general practices contributing to the QRESEARCH general practice database. An open cohort of patients registered with these practices during the 10-year study period (January 1, 1995, to July 1, 2005) was included. Cases all involved patients aged between 25 and 100 years with a first-ever record of the index cancer during the study period. The sample consisted of 40 441 incident cases of 6 cancers (breast, colon, rectal, gastroesophageal, prostate, and respiratory). The authors found up to 5 controls per case matched by age, sex, general practice, and calendar time. Odds ratios were computed for cancer risk association with schizophrenia and bipolar disorder, after adjusting for smoking, body mass index, socioeconomic status, comorbidities, and prescribed medications, including antipsychotics. After adjustment, patients with schizophrenia had increased colon cancer risk (adjusted OR, 2.90; 95% confidence interval [CI], 1.85-4.57), a marginal increased breast cancer risk (adjusted OR, 1.52; 95% CI, 1.10-2.11), and a decreased respiratory cancer risk (adjusted OR, 0.53, 95% CI, 0.34-0.85). Patients with bipolar disorder had cancer risks similar to patients with neither condition after adjustment. The authors concluded that patients with schizophrenia had a significantly higher risk of colon cancer and a lower risk of respiratory cancer compared with control subjects. In contrast, the risks of cancer in patients with and without bipolar disorder are similar. The authors observed a relatively small number of patients with schizophrenia among some of the subgroups of cancer cases. Which of the following has resulted from the low absolute prevalence of the risk factor among the exposed?

Table 1. Prevalence of Schizophrenia and Bipolar Disorder in Cases and Controls by Cancer Type

	Cancer					
	Breast	Colon	Rectal	Gastro-esophageal	Prostate	Respiratory
Cancer cases, No. (%)						
Schizophrenia only	47 (0.45)	31 (0.61)	8 (0.25)	12 (0.31)	12 (0.12)	20 (0.27)
Bipolar disorder only	42 (0.40)	12 (0.23)	6 (0.18)	7 (0.18)	15 (0.15)	32 (0.43)
Both conditions	2 (0.02)	1 (0.02)	0	2 (0.05)	1 (0.01)	3 (0.04)
Neither condition	10 444 (99.14)	5064 (99.14)	3234 (99.57)	3833 (99.46)	10 162 (99.73)	7451 (99.27)
Total	10 535	5108	3248	3854	10 190	7506
Cancer controls, No. (%)						
Schizophrenia only	143 (0.29)	48 (0.20)	48 (0.31)	48 (0.26)	108 (0.22)	139 (0.39)
Bipolar disorder only	154 (0.31)	55 (0.22)	30 (0.19)	36 (0.19)	85 (0.17)	94 (0.26)
Both conditions	10 (0.02)	3 (0.01)	2 (0.01)	5 (0.03)	8 (0.02)	9 (0.03)
Neither condition	49 767 (99.39)	24 352 (99.57)	15 472 (99.49)	18 388 (99.52)	48 547 (99.59)	35 739 (99.33)
Total	50 074	24 458	15 552	18 477	48 748	35 981

Select one:

- ☐ Spuriously positive results
- ☐ Need for a large sample size
- ☐ Smaller p values
- ☐ Errors due to multiple testing
- ☐ Inflated size of the effect

Your answer is incorrect.

When the prevalence of risk factor is very small, a large group of cases are required to detect meaningful differences in a case-control study.

The correct answer is: Need for a large sample size

Question **32**

Not answered

Marked out of 1

While studying the association of disease and a putative risk factor exposure rates, which of the following is a useful study design if the disease under investigation is rare even after the exposure?

Select one:

- ☐ Retrospective cohort study
- ☐ Prospective cohort study
- ☐ Cross sectional study
- ☐ Case control study
- ☐ Randomised trial

Your answer is incorrect.

A case-control study is the most useful design to study rare diseases. Cohort study may need a high number of subjects to detect the development of rare diseases.

The correct answer is: Case control study

Question 33

Not answered

Marked out of 1

Read the following abstract and answer the questions below. The following is extracted from an article on the association between schizophrenia and obstetric complications. Dalman et al, The British Journal of Psychiatry (2001), 179:403-8. Methods: From the Stockholm County In-Patient Register and community registers, the authors identified 524 individuals born in Stockholm since January 1960 who had been in-patients and had received a diagnosis of schizophrenia between January 1971 and June 1994 (n=648; of which 120 were excluded - Deaths (n=51), not resident in Stockholm county (n=32), birth records could not be retrieved (n=37)). For each case, 2 controls of same gender, resident in the same county, and born next in time from the same hospital were identified. As four cases did not have well-matched controls, they were excluded. In total 524 patients with schizophrenia and 1043 controls were included. Data on obstetric complications were obtained from birth records. APGAR scores for asphyxia were assigned by a midwife using the birth records, and confirmed by 2 paediatricians independently with excellent agreement between the two raters (? 0.83). In addition, the presence or absence of pre-eclampsia was also recorded in accordance with a predefined criteria. Other indicators of foetal growth impairment such as low birth weight, delay (>7 days) in gaining weight after birth, birth length and head circumference were also recorded. Indicators of hypoxia/ischaemia around birth such as emergency Caesarean section, placental abruption, cord prolapse, foetal heart rate <100 or >160 b.p.m., and prolonged hospital stay for the child were also noted. Odds ratios (ORs) and 95% confidence intervals (CIs) for schizophrenia in relation to individual obstetric complications were calculated for matched case--control sets using conditional logistic regression. Odds ratios were adjusted for possible confounding variables, both one by one and simultaneously. Results: The results are shown in the below table. Adjusting either separately or simultaneously for maternal history of psychotic illness, maternal age, parity, socio-economic classification, civil status and attendance at antenatal care did not substantially alter the odds ratios for most of the obstetric complications listed. However, after adjusting for these possible confounding variables, the odds ratio for schizophrenia in relation to signs of asphyxia at birth was increased to 4.5 (95% CI 2.2-9.5). The authors estimated a population-attributable fraction for birth asphyxia as 7.7% for schizophrenia using the data for population prevalence of birth asphyxia in Stockholm (not discussed in the study). Comparatively uncommon exposures such as pre-eclampsia, low birth-weight and short gestational age (<33 weeks) showed increased risk estimates but were not statistically significant. The strengths of associations between schizophrenia and all other complications were reduced and no longer statistically significant after the statistical adjustments. What type of study has been reported in the above paper?

Table 3 Adjusted odds ratios for schizophrenia

Complication	Odds ratio (95% CI)		
	Unadjusted	Adjusted for possible confounders ¹	Adjusted for other obstetric complications ²
Pre-eclampsia			
No	1.0	1.0	1.0
Yes	1.6 (0.7–3.5)	1.9 (0.8–5.0)	0.9 (0.3–3.0)
Birth weight (g)			
0–2499	1.8 (1.0–3.1)	1.4 (0.7–2.8)	0.7 (0.2–2.0)
2500–4499	1.0	1.0	1.0
4500+	0.5 (0.2–1.2)	0.6 (0.2–1.5)	0.5 (0.2–1.4)
Gestation (weeks)			
<33	2.7 (0.7–9.7)	2.2 (0.6–8.7)	1.1 (0.2–6.2)
37–42	1.0	1.0	1.0
Small for gestational age			
No	1.0	1.0	1.0
Yes	1.5 (0.9–2.4)	1.5 (0.9–2.6)	1.0 (0.5–2.0)
Delay in gaining weight after birth (days)			
0–7	1.0	1.0	1.0

Select one:

- ☐ Prospective cohort
- ☐ Retrospective cohort
- ☐ Cross-sectional study
- ☐ Ecological study
- ☐ Case control

8-20

Asphyxia at birth³

No signs of asphyxia

Signs of asphyxia

Child remained in hospital

1.9 (1.0–3.4)

1.6 (0.8–3.3)

1.3 (0.6–3.0)

1.0

1.0

1.0

2.7 (1.5–4.8)

4.5 (2.2–9.5)

4.4 (1.9–10.3)

Your answer is incorrect.

A case register and a community register were used to identify patients with schizophrenia (i.e. cases) and controls. This was followed by an estimation of exposure status (birth complications). The study reports odds ratios. This is a case control study.

The correct answer is: Case control

National In-Patient Register 1971–1996), maternal age (< 20, 20–24, 25–29, 30–34, 35–39, 40+), parity (1, 2–3, 4+), socio-economic classification (Group I i.e. professionals, Group II or III i.e. manual and non-manual employees), civil status (married, unmarried or divorced) and attendance to antenatal care (regular, none or irregular).

2. Adjusted for possible confounders and further adjusted simultaneously for all other complications listed in Table 3.

3. Signs of asphyxia based on complications expected to yield an Apgar score <7 at 1, 5 or 10 min.

Question **34**

Not answered

Marked out of 1

Read the following abstract and answer the questions below. The following is extracted from an article on the association between schizophrenia and obstetric complications. Dalman et al, The British Journal of Psychiatry (2001), 179:403-8. Methods: From the Stockholm County In-Patient Register and community registers, the authors identified 524 individuals born in Stockholm since January 1960 who had been in-patients and had received a diagnosis of schizophrenia between January 1971 and June 1994 (n=648; of which 120 were excluded - Deaths (n=51), not resident in Stockholm county (n=32), birth records could not be retrieved (n=37)). For each case, 2 controls of same gender, resident in the same county, and born next in time from the same hospital were identified. As four cases did not have well-matched controls, they were excluded. In total 524 patients with schizophrenia and 1043 controls were included. Data on obstetric complications were obtained from birth records. APGAR scores for asphyxia were assigned by a midwife using the birth records, and confirmed by 2 paediatricians independently with excellent agreement between the two raters (? 0.83). In addition, the presence or absence of pre-eclampsia was also recorded in accordance with a predefined criteria. Other indicators of foetal growth impairment such as low birth weight, delay (>7 days) in gaining weight after birth, birth length and head circumference were also recorded. Indicators of hypoxia/ischaemia around birth such as emergency Caesarean section, placental abruption, cord prolapse, foetal heart rate <100 or >160 b.p.m., and prolonged hospital stay for the child were also noted. Odds ratios (ORs) and 95% confidence intervals (CIs) for schizophrenia in relation to individual obstetric complications were calculated for matched case--control sets using conditional logistic regression. Odds ratios were adjusted for possible confounding variables, both one by one and simultaneously. Results: The results are shown in the below table. Adjusting either separately or simultaneously for maternal history of psychotic illness, maternal age, parity, socio-economic classification, civil status and attendance at antenatal care did not substantially alter the odds ratios for most of the obstetric complications listed. However, after adjusting for these possible confounding variables, the odds ratio for schizophrenia in relation to signs of asphyxia at birth was increased to 4.5 (95% CI 2.2-9.5). The authors estimated a population-attributable fraction for birth asphyxia as 7.7% for schizophrenia using the data for population prevalence of birth asphyxia in Stockholm (not discussed in the study). Comparatively uncommon exposures such as pre-eclampsia, low birth-weight and short gestational age (<33 weeks) showed increased risk estimates but were not statistically significant. The strengths of associations between schizophrenia and all other complications were reduced and no longer statistically significant after the statistical adjustments. Which of the following bias/error is least likely in this study?

Table 3 Adjusted odds ratios for schizophrenia

Complication	Odds ratio (95% CI)		
	Unadjusted	Adjusted for possible confounders ¹	Adjusted for other obstetric complications ²
Pre-eclampsia			
No	1.0	1.0	1.0
Yes	1.6 (0.7–3.5)	1.9 (0.8–5.0)	0.9 (0.3–3.0)
Birth weight (g)			
0–2499	1.8 (1.0–3.1)	1.4 (0.7–2.8)	0.7 (0.2–2.0)
2500–4499	1.0	1.0	1.0
4500+	0.5 (0.2–1.2)	0.6 (0.2–1.5)	0.5 (0.2–1.4)
Gestation (weeks)			
<33	2.7 (0.7–9.7)	2.2 (0.6–8.7)	1.1 (0.2–6.2)
37–42	1.0	1.0	1.0
Small for gestational age			
No	1.0	1.0	1.0
Yes	1.5 (0.9–2.4)	1.5 (0.9–2.6)	1.0 (0.5–2.0)
Delay in gaining weight after birth (days)			
0–7	1.0	1.0	1.0

Select one:

- ☐ Confounding bias
- ☐ Exposure identification bias
- ☐ Recall bias
- ☐ Exclusion bias
- ☐ Type 2 error

Asphyxia at birth	1.9 (1.0–3.4)	1.6 (0.8–3.3)	1.3 (0.6–3.0)
No signs of asphyxia	1.0	1.0	1.0
Signs of asphyxia	2.7 (1.5–4.8)	4.5 (2.2–9.5)	4.4 (1.9–10.3)
Child remained in hospital			

Your answer is incorrect.

Note that the authors identified people with schizophrenia from a population-based register, making the process relatively free from selection bias. The obstetric information was obtained using birth records (and not from maternal recall) thus reducing potential information bias such as recall bias.

The correct answer is: Recall bias

- socio-economic classification (Group I i.e. professionals, Group II or III i.e. manual and non-manual employees), civil status (married, unmarried or divorced) and attendance to antenatal care (regular, none or irregular).
2. Adjusted for possible confounders and further adjusted simultaneously for all other complications listed in Table 3.
3. Signs of asphyxia based on complications expected to yield an Apgar score <7 at 1, 5 or 10 min.

Question 35

Not answered

Marked out of 1

Read the following abstract and answer the questions below. The following is extracted from an article on the association between schizophrenia and obstetric complications. Dalman et al, The British Journal of Psychiatry (2001), 179:403-8. Methods: From the Stockholm County In-Patient Register and community registers, the authors identified 524 individuals born in Stockholm since January 1960 who had been in-patients and had received a diagnosis of schizophrenia between January 1971 and June 1994 (n=648; of which 120 were excluded - Deaths (n=51), not resident in Stockholm county (n=32), birth records could not be retrieved (n=37)). For each case, 2 controls of same gender, resident in the same county, and born next in time from the same hospital were identified. As four cases did not have well-matched controls, they were excluded. In total 524 patients with schizophrenia and 1043 controls were included. Data on obstetric complications were obtained from birth records. APGAR scores for asphyxia were assigned by a midwife using the birth records, and confirmed by 2 paediatricians independently with excellent agreement between the two raters (? 0.83). In addition, the presence or absence of pre-eclampsia was also recorded in accordance with a predefined criteria. Other indicators of foetal growth impairment such as low birth weight, delay (>7 days) in gaining weight after birth, birth length and head circumference were also recorded. Indicators of hypoxia/ischaemia around birth such as emergency Caesarean section, placental abruption, cord prolapse, foetal heart rate <100 or >160 b.p.m., and prolonged hospital stay for the child were also noted. Odds ratios (ORs) and 95% confidence intervals (CIs) for schizophrenia in relation to individual obstetric complications were calculated for matched case--control sets using conditional logistic regression. Odds ratios were adjusted for possible confounding variables, both one by one and simultaneously. Results: The results are shown in the below table. Adjusting either separately or simultaneously for maternal history of psychotic illness, maternal age, parity, socio-economic classification, civil status and attendance at antenatal care did not substantially alter the odds ratios for most of the obstetric complications listed. However, after adjusting for these possible confounding variables, the odds ratio for schizophrenia in relation to signs of asphyxia at birth was increased to 4.5 (95% CI 2.2-9.5). The authors estimated a population-attributable fraction for birth asphyxia as 7.7% for schizophrenia using the data for population prevalence of birth asphyxia in Stockholm (not discussed in the study). Comparatively uncommon exposures such as pre-eclampsia, low birth-weight and short gestational age (<33 weeks) showed increased risk estimates but were not statistically significant. The strengths of associations between schizophrenia and all other complications were reduced and no longer statistically significant after the statistical adjustments. Which of the following exposures showed a statistically significant association with schizophrenia before adjusting for potential confounders, but lost its significance upon adjustment?

Table 3 Adjusted odds ratios for schizophrenia

Complication	Odds ratio (95% CI)		
	Unadjusted	Adjusted for possible confounders ¹	Adjusted for other obstetric complications ²
Pre-eclampsia			
No	1.0	1.0	1.0
Yes	1.6 (0.7–3.5)	1.9 (0.8–5.0)	0.9 (0.3–3.0)
Birth weight (g)			
0–2499	1.8 (1.0–3.1)	1.4 (0.7–2.8)	0.7 (0.2–2.0)
2500–4499	1.0	1.0	1.0
4500+	0.5 (0.2–1.2)	0.6 (0.2–1.5)	0.5 (0.2–1.4)
Gestation (weeks)			
<33	2.7 (0.7–9.7)	2.2 (0.6–8.7)	1.1 (0.2–6.2)
37–42	1.0	1.0	1.0
Small for gestational age			
No	1.0	1.0	1.0
Yes	1.5 (0.9–2.4)	1.5 (0.9–2.6)	1.0 (0.5–2.0)
Delay in gaining weight after birth (days)			
0–7	1.0	1.0	1.0

Select one:

- ☐ Lower birth weight
- ☐ Pre-eclampsia
- ☐ None of the studied factors
- ☐ Greater delay in gaining weight
- ☐ Child remaining in the hospital

	1.7 (1.1–2.8)	1.5 (0.8–2.9)	1.7 (1.1–2.8)
Asphyxia at birth³	1.9 (1.0–3.4)	1.6 (0.8–3.3)	1.3 (0.6–3.0)
No signs of asphyxia	1.0	1.0	1.0
Signs of asphyxia	2.7 (1.5–4.8)	4.5 (2.2–9.5)	4.4 (1.9–10.3)
Child remained in hospital	1.7 (1.1–2.8)	1.5 (0.8–2.9)	1.7 (1.1–2.8)

Your answer is incorrect.

In the given table, 'Child remained in hospital' is associated with odds ratio 1.7 (1.1-2.8) before adjusting. Only when both confounding factors and all other independent variables are adjusted for, the OR drops to 1.5 and loses its significance (0.8 - 2.9). To satisfy the description in the question, an exposure must show significant effect before adjustment (i.e. OR must not touch the value of no-significance i.e. 1) AND must become non-significant on controlling for possible confounders (column 2 of the ORs). None of the variables satisfy this effect.

The correct answer is: Child remaining in the hospital

2. Adjusted for possible confounders and further adjusted simultaneously for all other complications listed in Table 3.
3. Signs of asphyxia based on complications expected to yield an Apgar score <7 at 1, 5 or 10 min.

Question 36

Not answered

Marked out of 1

Read the following abstract and answer the questions below. The following is extracted from an article on the association between schizophrenia and obstetric complications. Dalman et al, The British Journal of Psychiatry (2001), 179:403-8. Methods: From the Stockholm County In-Patient Register and community registers, the authors identified 524 individuals born in Stockholm since January 1960 who had been in-patients and had received a diagnosis of schizophrenia between January 1971 and June 1994 (n=648; of which 120 were excluded - Deaths (n=51), not resident in Stockholm county (n=32), birth records could not be retrieved (n=37)). For each case, 2 controls of same gender, resident in the same county, and born next in time from the same hospital were identified. As four cases did not have well-matched controls, they were excluded. In total 524 patients with schizophrenia and 1043 controls were included. Data on obstetric complications were obtained from birth records. APGAR scores for asphyxia were assigned by a midwife using the birth records, and confirmed by 2 paediatricians independently with excellent agreement between the two raters (κ 0.83). In addition, the presence or absence of pre-eclampsia was also recorded in accordance with a predefined criteria. Other indicators of foetal growth impairment such as low birth weight, delay (>7 days) in gaining weight after birth, birth length and head circumference were also recorded. Indicators of hypoxia/ischaemia around birth such as emergency Caesarean section, placental abruption, cord prolapse, foetal heart rate <100 or >160 b.p.m., and prolonged hospital stay for the child were also noted. Odds ratios (ORs) and 95% confidence intervals (CIs) for schizophrenia in relation to individual obstetric complications were calculated for matched case-control sets using conditional logistic regression. Odds ratios were adjusted for possible confounding variables, both one by one and simultaneously. Results: The results are shown in the below table. Adjusting either separately or simultaneously for maternal history of psychotic illness, maternal age, parity, socio-economic classification, civil status and attendance at antenatal care did not substantially alter the odds ratios for most of the obstetric complications listed. However, after adjusting for these possible confounding variables, the odds ratio for schizophrenia in relation to signs of asphyxia at birth was increased to 4.5 (95% CI 2.2-9.5). The authors estimated a population-attributable fraction for birth asphyxia as 7.7% for schizophrenia using the data for population prevalence of birth asphyxia in Stockholm (not discussed in the study). Comparatively uncommon exposures such as pre-eclampsia, low birth-weight and short gestational age (<33 weeks) showed increased risk estimates but were not statistically significant. The strengths of associations between schizophrenia and all other complications were reduced and no longer statistically significant after the statistical adjustments. Which of the following statement is correct with regard to the results of this study?

Table 3 Adjusted odds ratios for schizophrenia

Complication	Odds ratio (95% CI)		
	Unadjusted	Adjusted for possible confounders ¹	Adjusted for other obstetric complications ²
Pre-eclampsia			
No	1.0	1.0	1.0
Yes	1.6 (0.7–3.5)	1.9 (0.8–5.0)	0.9 (0.3–3.0)
Birth weight (g)			
0–2499	1.8 (1.0–3.1)	1.4 (0.7–2.8)	0.7 (0.2–2.0)
2500–4499	1.0	1.0	1.0
4500+	0.5 (0.2–1.2)	0.6 (0.2–1.5)	0.5 (0.2–1.4)
Gestation (weeks)			
<33	2.7 (0.7–9.7)	2.2 (0.6–8.7)	1.1 (0.2–6.2)
37–42	1.0	1.0	1.0
Small for gestational age			
No	1.0	1.0	1.0
Yes	1.5 (0.9–2.4)	1.5 (0.9–2.6)	1.0 (0.5–2.0)
Delay in gaining weight after birth (days)			
0–7	1.0	1.0	1.0

Select one:

- ☐ Pre-eclampsia is associated with schizophrenia, but the significance of this association is explained by other confounders
- ☐ Hospital admission of the child following birth is associated with schizophrenia, and the significance of this association is dependent on other obstetric complications
- ☐ Pre-eclampsia is independently associated with schizophrenia
- ☐ Hospital admission of the child following birth is associated with schizophrenia, but the significance of this association is explained by other confounders
- ☐ Hospital admission of the child following birth is independently associated with schizophrenia

Your answer is incorrect.

Pre-eclampsia, both in the unadjusted OR column and the adjusted column, has no statistical significance (the confidence intervals cross the value 1). Hospital admission of the child following birth is associated with schizophrenia even after adjusting for confounders but loses its significance when adjusting for other obstetric risk factors, hence the statement independent association with schizophrenia is untenable.

The correct answer is: Hospital admission of the child following birth is associated with schizophrenia, and the significance of this association is dependent on other obstetric complications

3. Signs of asphyxia based on complications expected to yield an Apgar score <7 at 1, 5 or 10 min.

Question 37

Not answered

Marked out of 1

Read the following abstract and answer the questions below. The following is extracted from an article on the association between schizophrenia and obstetric complications. Dalman et al, The British Journal of Psychiatry (2001), 179:403-8. Methods: From the Stockholm County In-Patient Register and community registers, the authors identified 524 individuals born in Stockholm since January 1960 who had been in-patients and had received a diagnosis of schizophrenia between January 1971 and June 1994 (n=648; of which 120 were excluded - Deaths (n=51), not resident in Stockholm county (n=32), birth records could not be retrieved (n=37)). For each case, 2 controls of same gender, resident in the same county, and born next in time from the same hospital were identified. As four cases did not have well-matched controls, they were excluded. In total 524 patients with schizophrenia and 1043 controls were included. Data on obstetric complications were obtained from birth records. APGAR scores for asphyxia were assigned by a midwife using the birth records, and confirmed by 2 paediatricians independently with excellent agreement between the two raters (? 0.83). In addition, the presence or absence of pre-eclampsia was also recorded in accordance with a predefined criteria. Other indicators of foetal growth impairment such as low birth weight, delay (>7 days) in gaining weight after birth, birth length and head circumference were also recorded. Indicators of hypoxia/ischaemia around birth such as emergency Caesarean section, placental abruption, cord prolapse, foetal heart rate <100 or >160 b.p.m., and prolonged hospital stay for the child were also noted. Odds ratios (ORs) and 95% confidence intervals (CIs) for schizophrenia in relation to individual obstetric complications were calculated for matched case--control sets using conditional logistic regression. Odds ratios were adjusted for possible confounding variables, both one by one and simultaneously. Results: The results are shown in the below table. Adjusting either separately or simultaneously for maternal history of psychotic illness, maternal age, parity, socio-economic classification, civil status and attendance at antenatal care did not substantially alter the odds ratios for most of the obstetric complications listed. However, after adjusting for these possible confounding variables, the odds ratio for schizophrenia in relation to signs of asphyxia at birth was increased to 4.5 (95% CI 2.2-9.5). The authors estimated a population-attributable fraction for birth asphyxia as 7.7% for schizophrenia using the data for population prevalence of birth asphyxia in Stockholm (not discussed in the study). Comparatively uncommon exposures such as pre-eclampsia, low birth-weight and short gestational age (<33 weeks) showed increased risk estimates but were not statistically significant. The strengths of associations between schizophrenia and all other complications were reduced and no longer statistically significant after the statistical adjustments. In this study, preeclampsia in the mother did not emerge as a significant factor associated with schizophrenia in the child at later life, though many previous studies have found a significant relationship. Which of the following is most likely to explain this discrepancy?

Table 3 Adjusted odds ratios for schizophrenia

Complication	Odds ratio (95% CI)		
	Unadjusted	Adjusted for possible confounders ¹	Adjusted for other obstetric complications ²
Pre-eclampsia			
No	1.0	1.0	1.0
Yes	1.6 (0.7–3.5)	1.9 (0.8–5.0)	0.9 (0.3–3.0)
Birth weight (g)			
0–2499	1.8 (1.0–3.1)	1.4 (0.7–2.8)	0.7 (0.2–2.0)
2500–4499	1.0	1.0	1.0
4500+	0.5 (0.2–1.2)	0.6 (0.2–1.5)	0.5 (0.2–1.4)
Gestation (weeks)			
<33	2.7 (0.7–9.7)	2.2 (0.6–8.7)	1.1 (0.2–6.2)
37–42	1.0	1.0	1.0
Small for gestational age			
No	1.0	1.0	1.0
Yes	1.5 (0.9–2.4)	1.5 (0.9–2.6)	1.0 (0.5–2.0)
Delay in gaining weight after birth (days)			
0–7	1.0	1.0	1.0

Select one:

- ☐ Presence of confounding bias in the current study.
- ☐ Over-matching of controls
- ☐ Type 1 error
- ☐ There is no true association between pre-eclampsia and schizophrenia
- ☐ Type 2 error

8-20

Asphyxia at birth

No signs of asphyxia

Signs of asphyxia

Child remained in hospital

1.9 (1.0–3.4)

1.0

2.7 (1.5–4.8)

1.6 (0.8–3.3)

1.0

4.5 (2.2–9.5)

1.3 (0.6–3.0)

1.0

4.4 (1.9–10.3)

Your answer is incorrect.

The power to detect an association between a risk factor and outcome in case-control studies depends very much on the proportion of subjects exposed to the putative risk factor in the sample. For example, if only one patient with schizophrenia and 2 controls have a history of childhood sexual abuse (CSA) in a total sample of over 1000, the association between the CSA and schizophrenia will not be significant and larger samples may be required to detect a significant effect. This is a type 2 error. The careful case-control matching together with statistical adjustment for confounders has significantly reduced potential bias.

The correct answer is: Type 2 error

2. Adjusted for possible confounders and further adjusted simultaneously for all other complications listed in Table 3.

3. Signs of asphyxia based on complications expected to yield an Apgar score <7 at 1, 5 or 10 min.

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